Project: A molecular dynamics simulation of **binary mixture of Lennard-Jones particles.**

System: Mixture of Ar-Kr

Objective: Create a molecular dynamics (MD) code able to model a binary mixture of Lennard-Jones (LJ) particles (types A,B), as argon-krypton.

Outline:

Modify the MD code ‘leapfroglj.f’ and the input files in order to create a new version able to model a binary mixture of LJ liquids. You'll need to create a new initial setup (leap-conf-new.data) containing all initial coordinates and velocities. HINT: Use a equally placed cubic geometry and a Maxwell-Boltzmann classical velocity distribution.

1. As usual, plot temperature versus time and energy vs. time. Make sure your NVE simulation is fully equilibrated (fluctuations below 5%). In the case that fluctuations are too big, you can introduce a (crude) velocity correction (every N time steps) in the way of a Berendsen's thermal bath to control the temperature.

2. Obtain radial distribution functions g(r) of AA, BB and AB and discuss the results.

3. Compute running coordination numbers n(r) via integration of the g(r)'s. Discuss your results.

4. Calculate velocity autocorrelation functions (VACF): $\langle v_A(0)\cdot v_A(t)\rangle$ and $\langle v_B(0)\cdot v_B(t)\rangle$. Discuss the results.

5. Compute mean square displacements $|r_A(t)-r_A(0)|^2$ and $|r_B(t)-r_B(0)|^2$. Then from Einstein's diffusion formula, obtain the corresponding self-diffusion coefficients D_A, D_B. Discuss the results.

Lennard-Jones interaction potentials

atom	mass(uma)	ε/k_B (K)	σ(Å)
Ar	39,948	119,18	3,405
Kr	83,798	167,00	3,670

Use the **Lorentz-Berthelot** rules for cross interactions:

- Use σ_{AB} and ε_{AB} for the reduced units

$$\sigma_{AB} = \frac{1}{2}(\sigma_A + \sigma_B)$$

- Consider the following state points:

$$\varepsilon_{AB} = \sqrt{\varepsilon_A \varepsilon_B}$$

T*= 1.00, ρ*=0.71,
T*= 1.00, ρ*=0.86

Partial molar fractions: x_{Ar} = 0.485 ; x_{Kr} = 0.515

Reduced units for mass:

$$\mu = \chi_{Ar} m_{Ar} + \chi_{Kr} m_{Kr}$$

Some results to compare with:

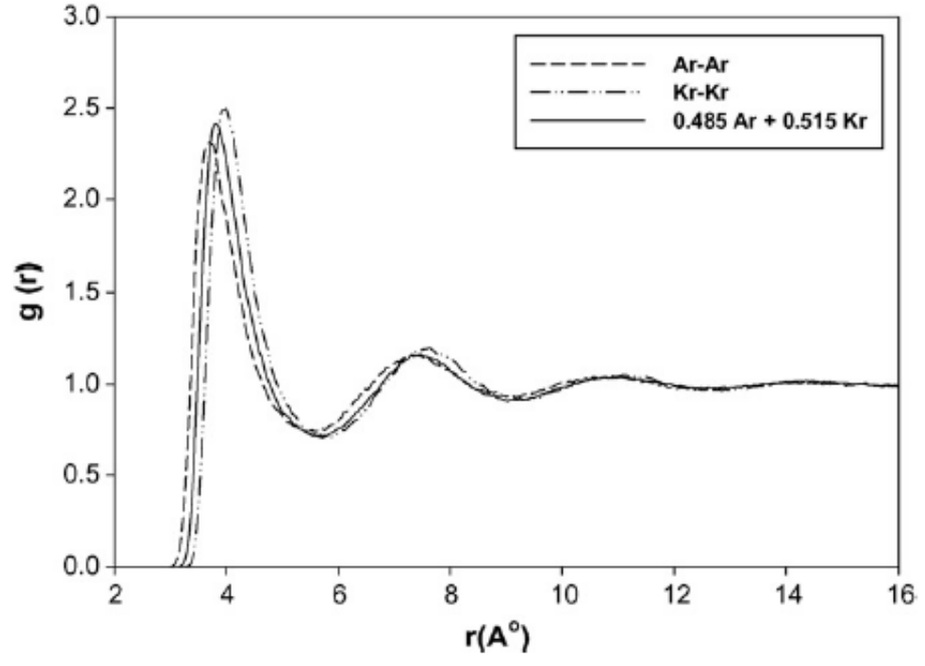
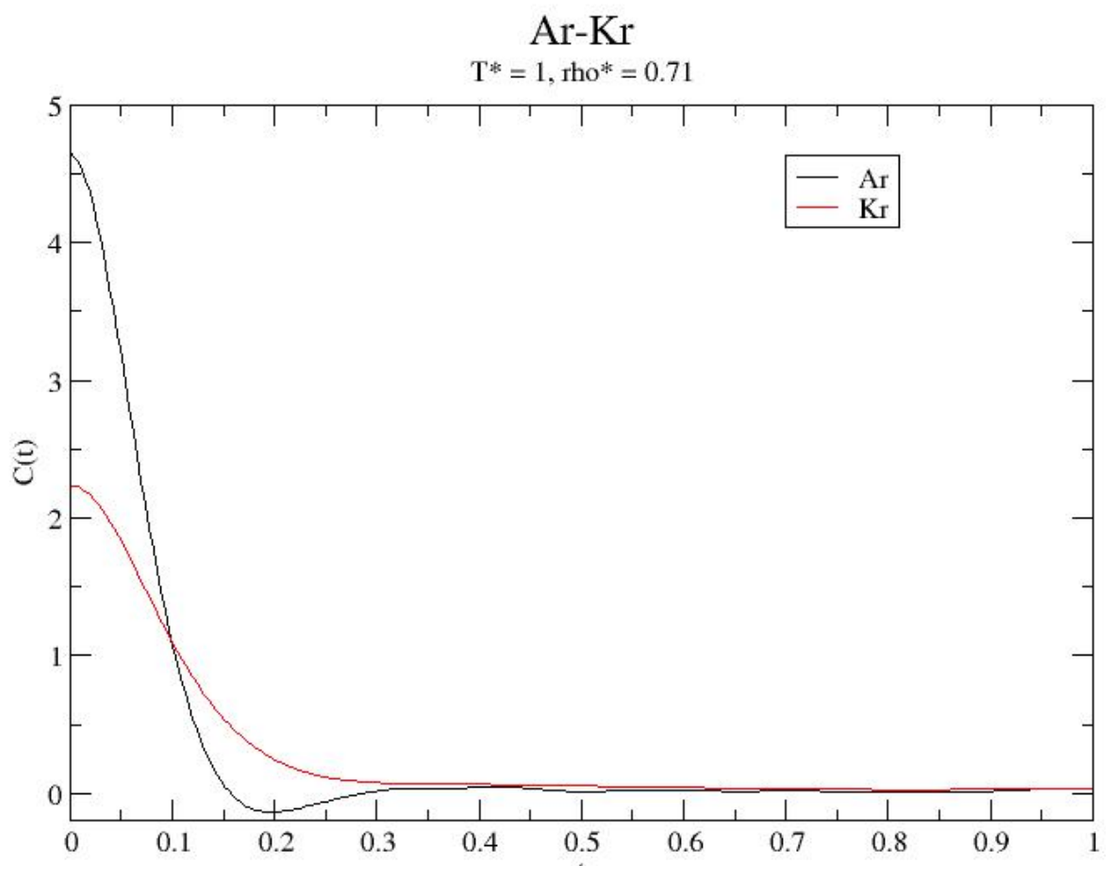
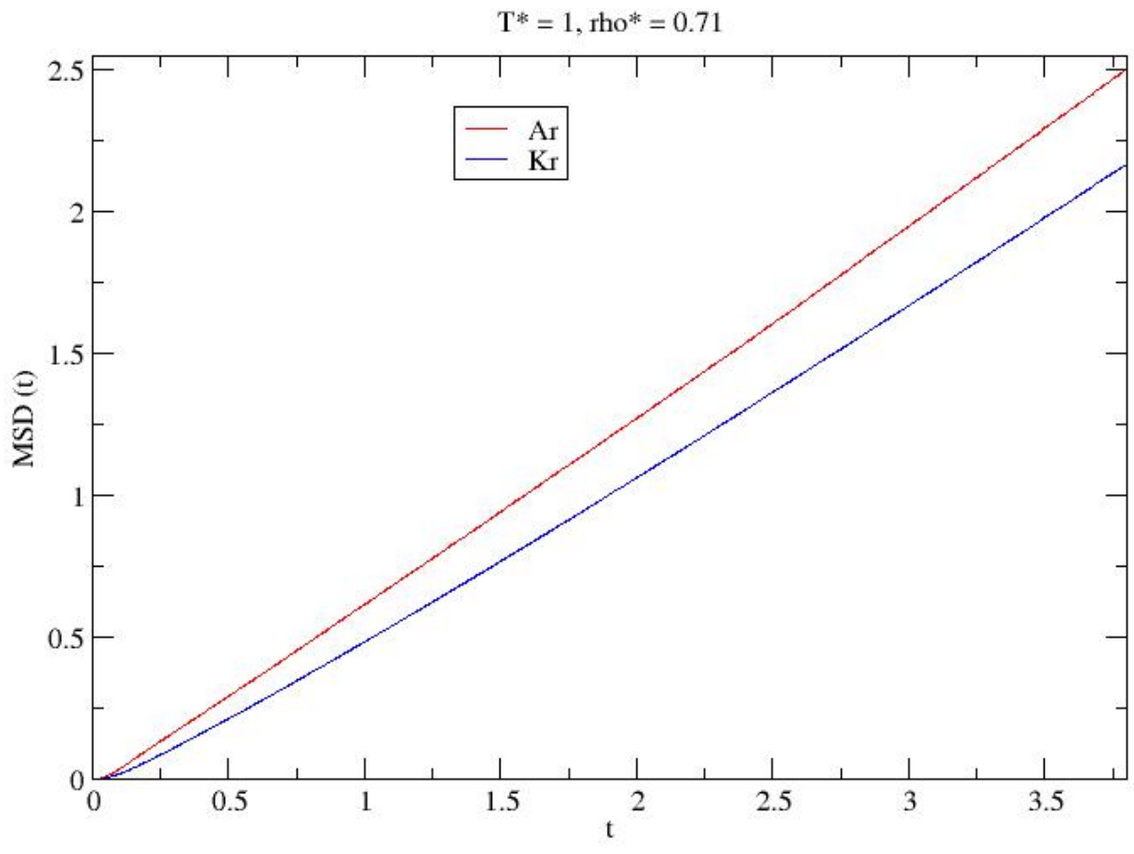


Fig. 4. RDFs of Ar–Ar, Kr–Kr, and 0.485Ar + 0.515Kr at T^* = 1.00877 (142.68 K) and ρ^* = 0.71205 calculated using Lennard–Jones potential.


Ar-Kr
T* = 1, rho* = 0.71
C(t)
Ar
Kr
0
1
2
3
4
5
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1


Ar-Kr
T* = 1, rho* = 0.71
MSD (t)
t
Ar
Kr
0
0.5
1
1.5
2
2.5
0
0.5
1
1.5
2
2.5
3
3.5

Indications:

1. To compute g(r)s

$$g_{AA}(r) = \frac{1}{\rho\chi_A}\frac{n_{AA}(r)}{4\pi r^2 dr} = \frac{n_{AA}(r)}{n_{id,A}(r)}$$

$$g_{BB}(r) = \frac{1}{\rho\chi_B}\frac{n_{BB}(r)}{4\pi r^2 dr} = \frac{n_{BB}(r)}{n_{id,B}(r)}$$

$$g_{AB}(r) = \frac{1}{\rho}\frac{n_{AB}(r)}{4\pi r^2 dr} = \frac{n_{AB}(r)}{n_{id}(r)}$$

2. To consider reduced units for energies (total energy = sum of energies)

$$U^*_{ArKr} = 4\left[\left(\frac{1}{r^*}\right)^{12} - \left(\frac{1}{r^*}\right)^{6}\right]$$

$$U^*_{ArAr} = 4\frac{\epsilon_{Ar}}{\epsilon_{ArKr}}\left[\left(\frac{\sigma_{Ar}}{\sigma_{ArKr}}\frac{1}{r^*}\right)^{12} - \left(\frac{\sigma_{Ar}}{\sigma_{ArKr}}\frac{1}{r^*}\right)^{6}\right]$$

(id. for KrKr)

3. To compute tail contributions

U^{tail} will have 3 contributions (A,B = Ar, Kr), computed by:

$$U_{AB}^{tail} = 2\pi N_A \rho_B \int_{r_c}^{\infty} \sum_{A,B} N_A N_B U_{AB}(r)\ r^2\ \mathrm{d}r$$

P^{tail} will have 3 contributions computed in analogous way:

$$P_{AB}^{tail} = -\frac{2\pi}{3V} \int_{r_c}^{\infty} \sum_{A,B} N_A N_B \frac{dU_{AB}(r)}{dr}\ r^3\ dr$$

Questions:

1. How the choice of the system size influences the performance of the simulations (CPU time, energy and temperature stability, etc.)?

2. Make Fourier transforms of the VACF and obtain spectral densities. What are the main vibrational motions in the system? How this is different of the mono-component system?

3. From the Lennard-Jones parameters for Ar and Kr:

   - Estimate the temperature of Ar and Kr at $T^* \equiv k_B T/\epsilon = 2.0$
   - Obtain the physical time corresponding to the time step of $\Delta t^* = 0.001$